ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 11

CIVIL TECHNOLOGY

MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 11 Civil Technology practice examination.

Written by Academic Central for practice, following the CAPS curriculum for Grade 11. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Penalise once per question for missing units. Accept other correct answers and small rounding differences.

QUESTION 1

1 1.1 A ✓✓ 1.2 A ✓✓ 1.3 B ✓✓ 1.4 A ✓✓ 1.5 A ✓✓ [10]

QUESTION 2

2.1 Any THREE: inspected by a competent person before use (daily) ✓; base plates on firm, level ground ✓; guardrails and toe boards on working platforms ✓; braced and tied to the building; never overloaded; safe ladder access (3)

2.2 (a) To set out and transfer levels (heights) on a building site ✓✓, e.g. for foundations and floors. (4)
(b) To compact fresh concrete ✓ by shaking out trapped air bubbles, giving dense, strong concrete ✓.

2.3 A person who has the knowledge, training and experience for the specific work ✓ and who knows the Act and its regulations that apply to it ✓. (2)

2.4 Any THREE: shore (support) the sides or cut them back to a safe slope ✓; keep spoil and machines at least 1 m from the edge ✓; provide a ladder for getting in and out ✓; barricade the trench; inspect it daily and after rain (3)

2.5 A disc rated lower than the grinder's speed spins faster than it was designed for ✓; it can shatter (burst) ✓ and the flying pieces can cause serious injury ✓. (3)

[15]

QUESTION 3

3.1.1 $\text{Wet volume} = 5 \times 4 \times 0{,}15$ ✓ $= 3\ \text{m}^3$ ✓ (3)
$\text{Dry volume} = 3 \times 1{,}5 = 4{,}5\ \text{m}^3$ ✓

3.1.2 $\text{Total parts} = 1 + 2 + 4 = 7$ (3)
$\text{Cement} = 4{,}5 \times 1/7 = 0{,}64\ \text{m}^3$ ✓ $\text{Sand} = 4{,}5 \times 2/7 = 1{,}29\ \text{m}^3$ ✓ $\text{Stone} = 4{,}5 \times 4/7 = 2{,}57\ \text{m}^3$ ✓

3.1.3 $\text{Mass} = 0{,}643 \times 1\,440 = 925{,}71\ \text{kg}$ ✓ (3)
$\text{Bags} = 925{,}71 \div 50 = 18{,}51$ ✓ $\rightarrow 19$ bags ✓ (must round up)

3.2 Place the clean, damp cone on a flat, non-absorbent base and stand on the foot rests ✓. Fill it in three equal layers ✓, rodding each layer 25 times with the tamping rod ✓. Strike off the top level and lift the cone slowly and vertically ✓. Measure the drop from the top of the cone to the highest point of the concrete: this is the slump ✓. (5)

3.3 Any TWO: lower strength ✓; more shrinkage and cracking ✓; porous, less durable concrete; segregation and bleeding (2)

3.4 Any TWO with purpose: accelerator ✓ – speeds up setting and early strength (cold weather) ✓; retarder ✓ – slows setting (hot weather, long transport) ✓; plasticiser – improves workability without extra water; air-entraining agent – resists frost damage; waterproofing agent – reduces water penetration (4)

[20]

QUESTION 4

4.1 (6)

$\text{Moments about A (clockwise} = \text{anticlockwise)}:$

$R_B \times 6$ ✓ $= (10 \times 1{,}5) + (15 \times 3) + (20 \times 4{,}5)$ ✓ $= 150\ \text{kN·m}$ ✓

$R_B = 25\ \text{kN}$ ✓

$\text{Upward forces} = \text{downward forces: } R_A = 45 - 25$ ✓ $= 20\ \text{kN}$ ✓

4.2 (6)

$\text{Flange: } A = 120 \times 20 = 2\,400\ \text{mm}^2$ ✓, $y = 100 + 10 = 110\ \text{mm}$ ✓

$\text{Web: } A = 20 \times 100 = 2\,000\ \text{mm}^2$ ✓, $y = 50\ \text{mm}$ ✓

$\bar{y} = (2\,400 \times 110 + 2\,000 \times 50) \div 4\,400$ ✓ $= 364\,000 \div 4\,400 = 82{,}73\ \text{mm}$ from the bottom ✓

4.3 (7)

$A = \pi \times 20^2 \div 4 = 314{,}16\ \text{mm}^2$ ✓

$\text{(a) } \sigma = F \div A = 50\,000 \div 314{,}16$ ✓ $= 159{,}15\ \text{MPa}$ ✓

$\text{(b) } \varepsilon = \sigma \div E = 159{,}15 \div 200\,000$ ✓ $= 7{,}96 \times 10^{-4}\ (0{,}000\,796)$ ✓

$\text{(c) } \Delta L = \varepsilon \times L = 7{,}96 \times 10^{-4} \times 2\,000$ ✓ $= 1{,}59\ \text{mm}$ ✓

4.4 (6)

(a) Stress is the internal resisting force per unit area ✓, σ = F ÷ A ✓.

(b) Strain is the change in length divided by the original length ✓, ε = ΔL ÷ L (no units) ✓.

(c) Within the elastic limit ✓, stress is directly proportional to strain ✓.

[25]

QUESTION 5

5.1 Any FOUR: strong and rigid enough to carry the wet concrete without bulging ✓; watertight joints so cement paste does not leak ✓; easy to erect and strip ✓; can be re-used ✓; accurate in shape, size and level; smooth inside face; economical (4)

5.2 The side formwork carries no load and can be removed after a day or two ✓. The soffit carries the full weight of the beam until the concrete has gained enough strength (about 14 to 21 days) ✓. (2)

5.3 Any FOUR: tie beam ✓; principal rafters ✓; king post ✓; struts ✓; ridge; purlins; wall plate (4)

5.4 A strip foundation is a continuous strip of concrete under the load-bearing walls ✓. A raft foundation is a reinforced concrete slab under the whole building ✓. A raft is used where the soil has a low bearing capacity ✓ or where uneven settlement is expected, e.g. clay or made-up ground ✓. (4)

5.5 Any TWO: stops rain penetrating to the inside wall ✓; better heat and sound insulation ✓; less condensation. Wall ties tie the two leaves together so the wall is stable ✓. (3)

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5.6 (a) $2\,700 \div 180 = 15$ risers ✓ (3)

(b) $2 \times 180 + G = 600$ ✓ $\rightarrow G = 240$ mm ✓

[20]

QUESTION 6

6.1 Floor area $= 4{,}2 \times 3{,}6 = 15{,}12\ m^2$ ✓; tile area $= 0{,}4 \times 0{,}4 = 0{,}16\ m^2$ ✓ (5)

$15{,}12 \div 0{,}16 = 94{,}5 \rightarrow 95$ tiles ✓

$95 \times 1{,}1 = 104{,}5$ ✓ $\rightarrow 105$ tiles ✓ (accept 104 if 10% was added before rounding)

6.2 Any FIVE: title of the drawing ✓; name of the client or project ✓; scale ✓; drawing number ✓; date ✓; name of the draughtsperson; site address; checked / approved by; sheet number (5)

[10]

TOTAL: 100

END OF MARKING GUIDELINE